Yahoo Stock Price Predictor

Raw Data

Date	High	Low	Open	Close	Volume	Adj Close
2015-11-23 00:00:00	2,095.6101	2,081.3899	2,089.4099	2,086.5901	3,587,980,000	2,086.5901
2015-11-24 00:00:00	2,094.1201	2,070.29	2,084.4199	2,089.1399	3,884,930,000	2,089.1399
2015-11-25 00:00:00	2,093	2,086.3	2,089.3	2,088.8701	2,852,940,000	2,088.8701
2015-11-26 00:00:00	2,093	2,086.3	2,089.3	2,088.8701	2,852,940,000	2,088.8701
2015-11-27 00:00:00	2,093.29	2,084.1299	2,088.8201	2,090.1101	1,466,840,000	2,090.1101

Descriptive Statistics

	High	Low	Open	Close	Volume	Adj Close
count	1,825	1,825	1,825	1,825	1,825	1,825
mean	2,660.7187	2,632.8176	2,647.7048	2,647.8563	3,869,626,822.9753	2,647.8563
std	409.6809	404.3101	407.17	407.3012	1,087,592,837.8897	407.3012
min	1,847	1,810.1	1,833.4	1,829.08	1,296,540,000	1,829.08
25%	2,348.3501	2,322.25	2,341.98	2,328.95	3,257,950,000	2,328.95
50%	2,696.25	2,667.8401	2,685.49	2,683.3401	3,609,740,000	2,683.3401
75%	2,930.79	2,900.71	2,913.8601	2,917.52	4,142,850,000	2,917.52
max	3,645.99	3,600.1599	3,612.0901	3,626.9099	9,044,690,000	3,626.9099

Rolling Mean and Standard Deviation of High Prices

	mean	std
0	2,019.3543	71.041
1	2,147.7403	36.2876
2	2,322.9696	60.2284
3	2,489.7066	55.8781
4	2,711.2537	63.1259
5	2,816.0889	68.3517
6	2,754.1656	131.7924
7	2,965.1531	74.0593
8	3,045.6693	267.6954
9	3,309.0766	165.7338

Augmented Dickey-Fuller Test for Stationarity

ADF Statistic: -0.713598

p-value: 0.843196

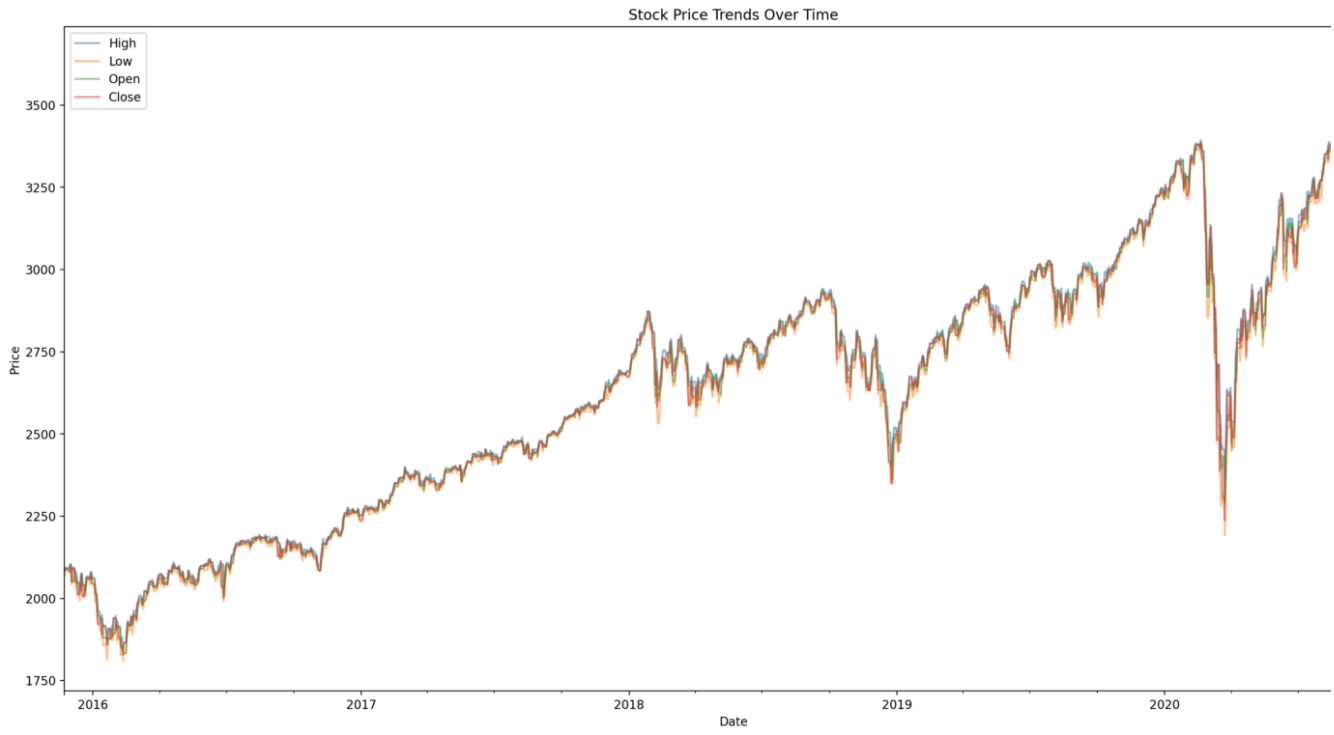
Critical Values:

1%: -3.434

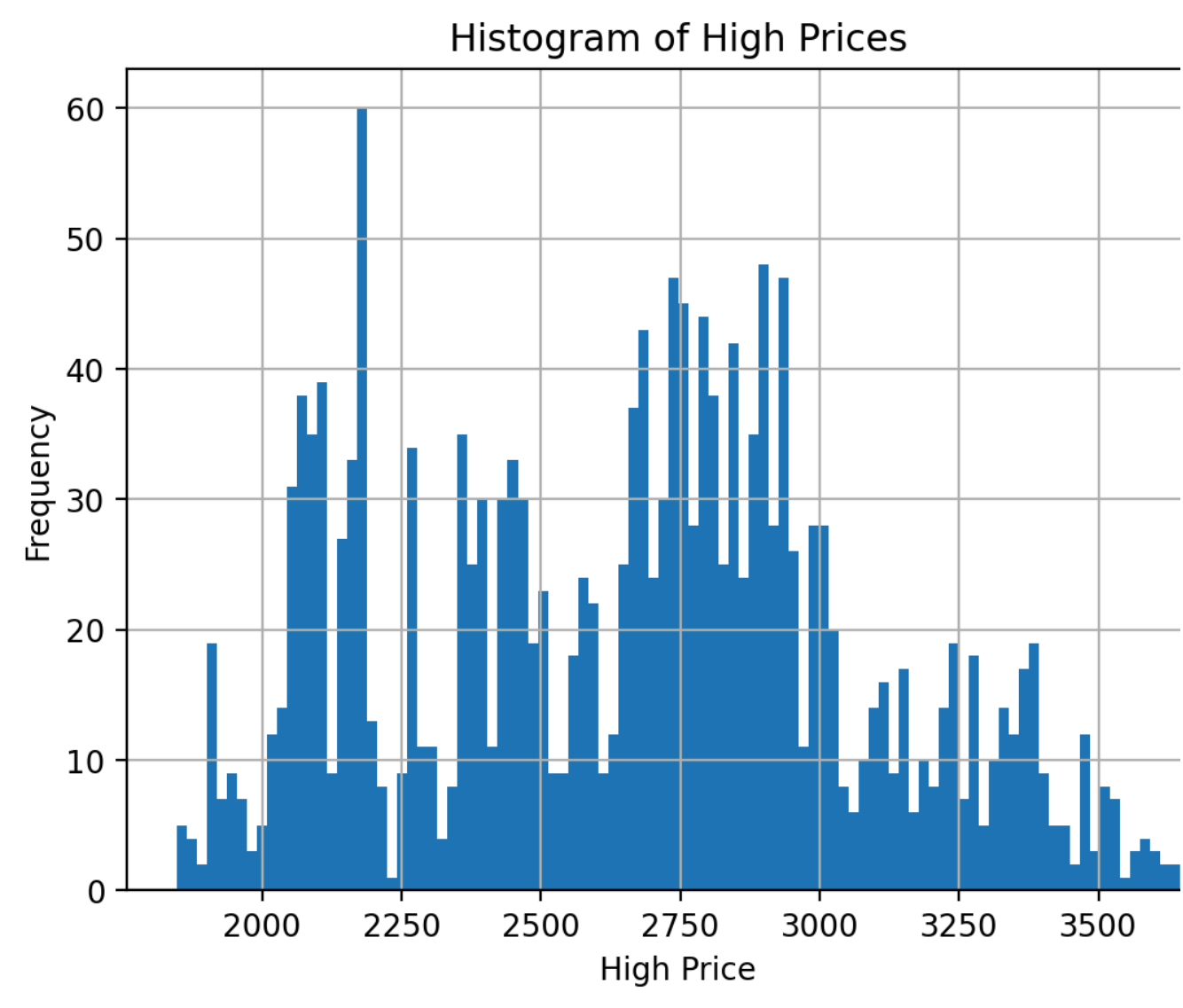
5%: -2.863

10%: -2.568

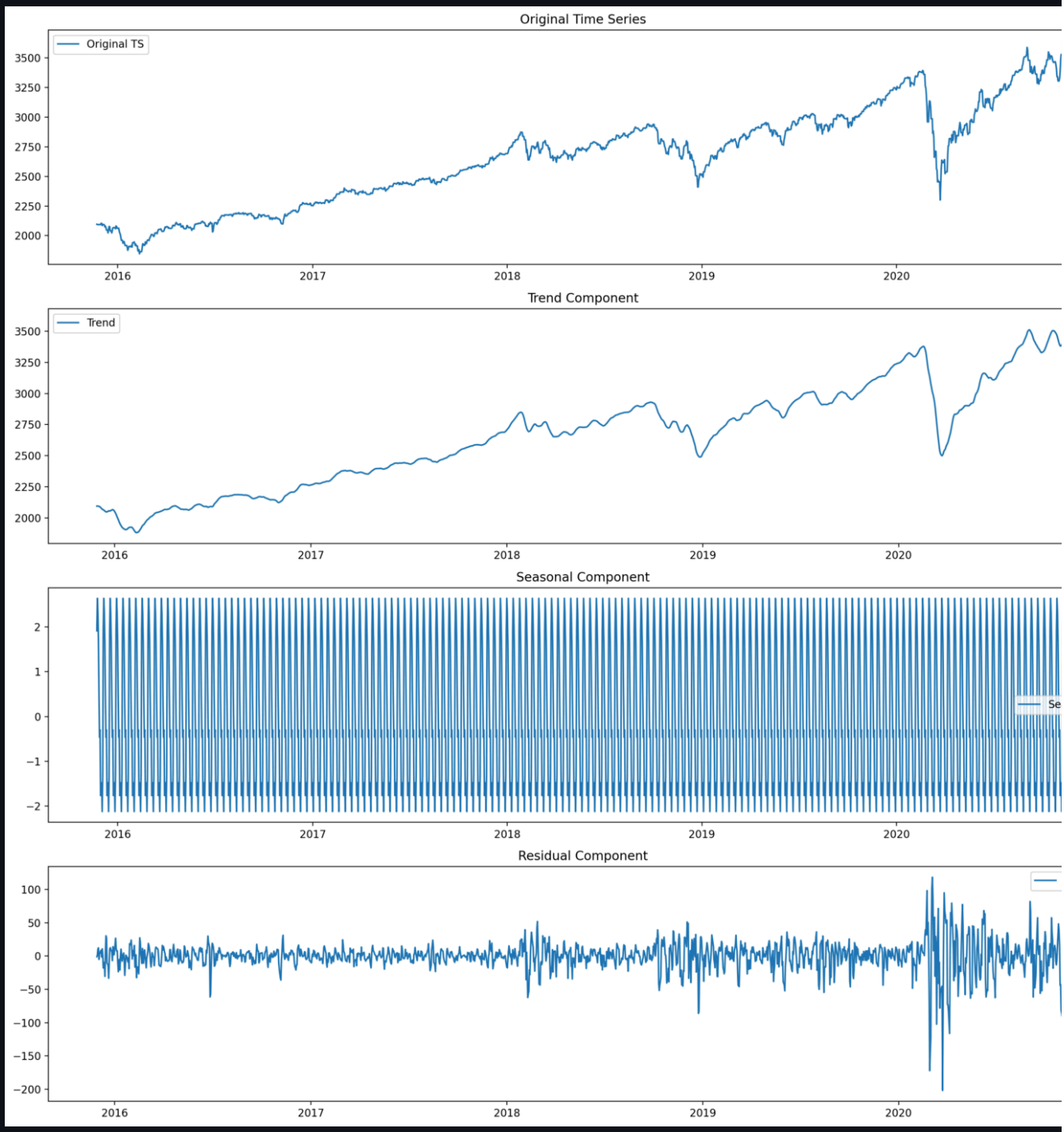
Stock Price Trends



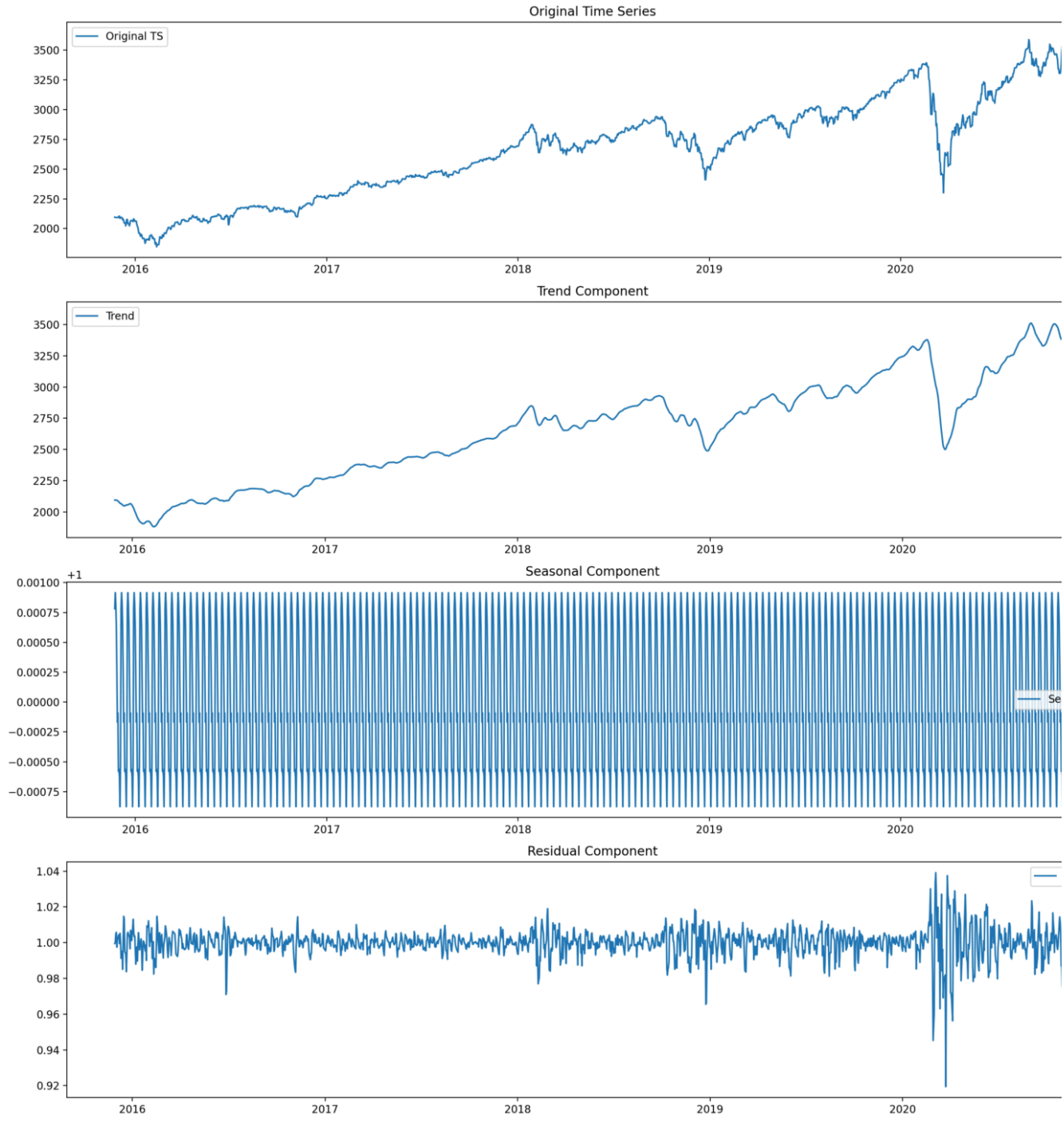
Distribution of High Prices



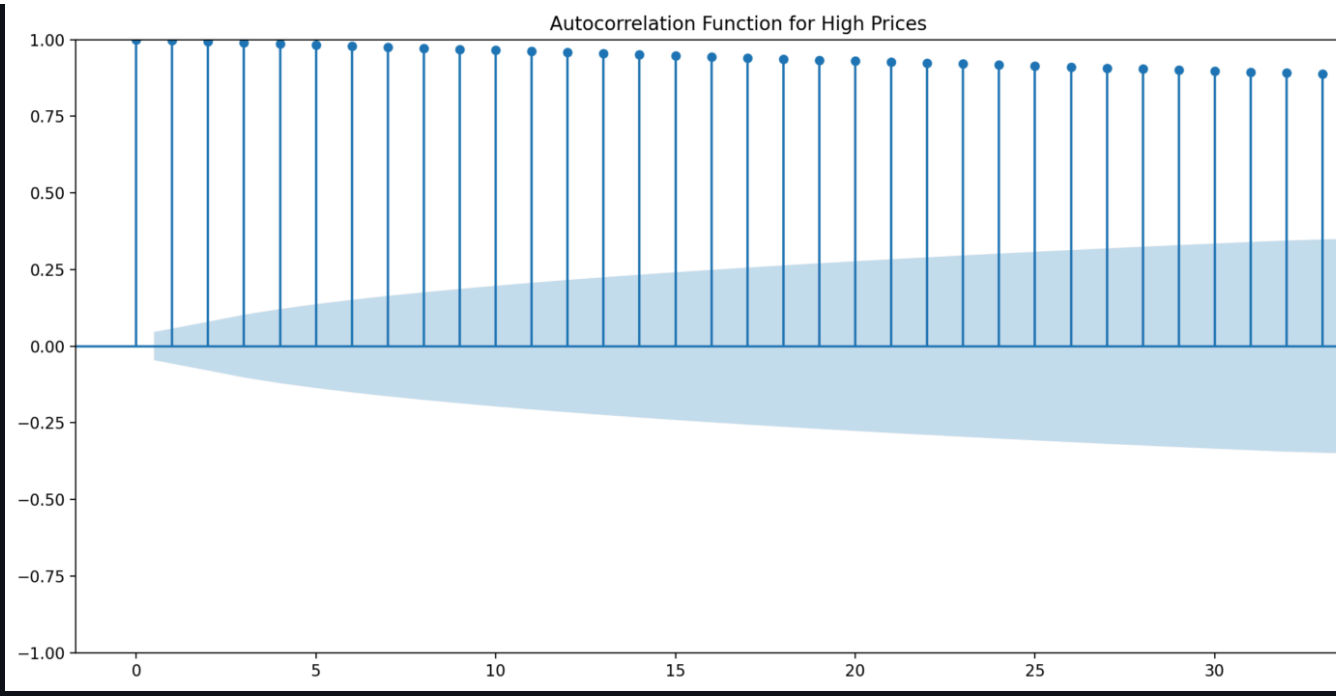
Seasonal Decomposition (Additive Model)



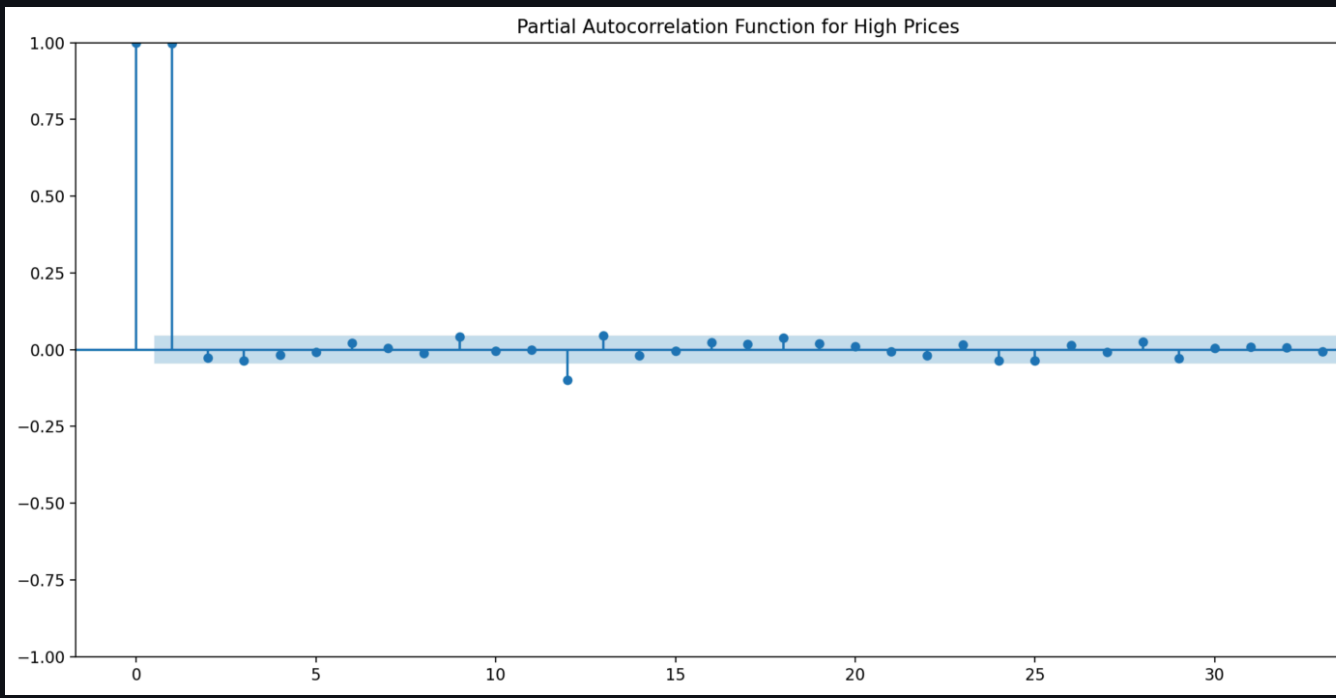
Seasonal Decomposition (Multiplicative Model)



Autocorrelation Function (ACF)



Partial Autocorrelation Function (PACF)



LSTM Model Training and Evaluation (Window=10)

Training data length: 1457

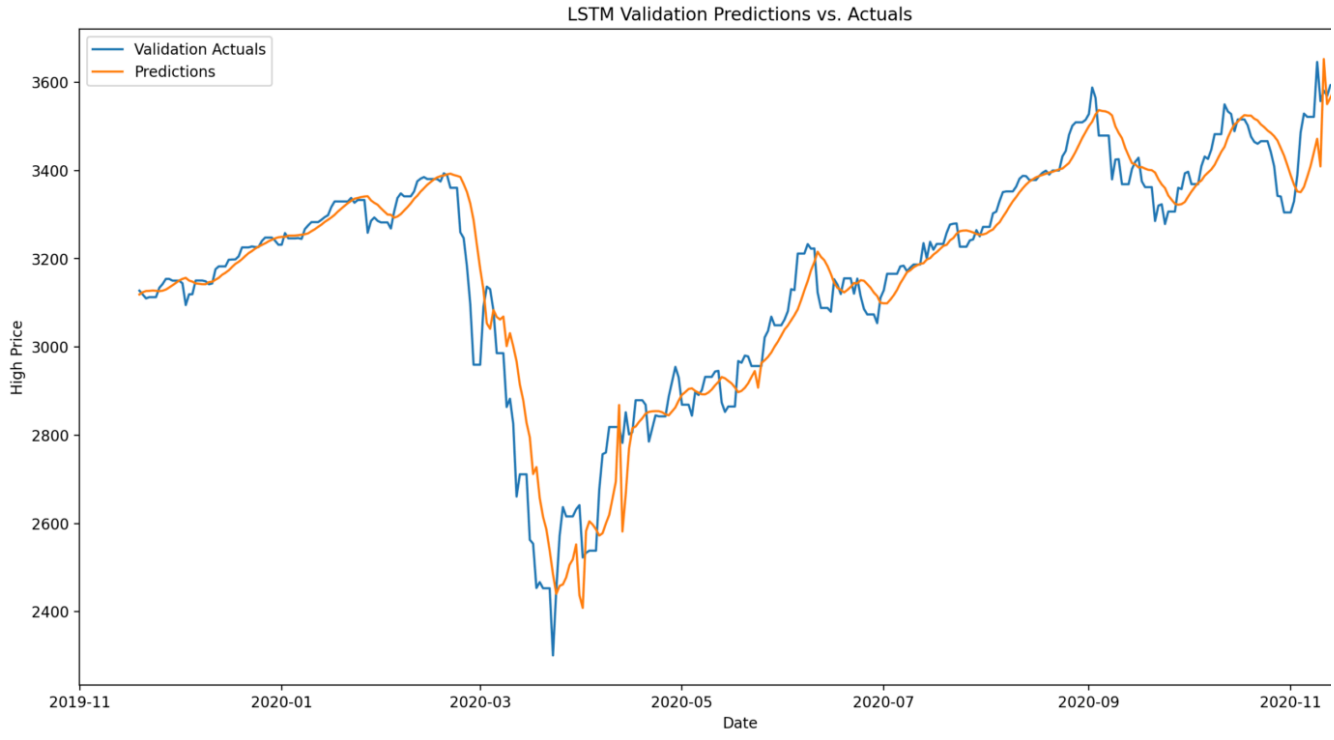
Window size for LSTM: 10

Initial LSTM model trained!

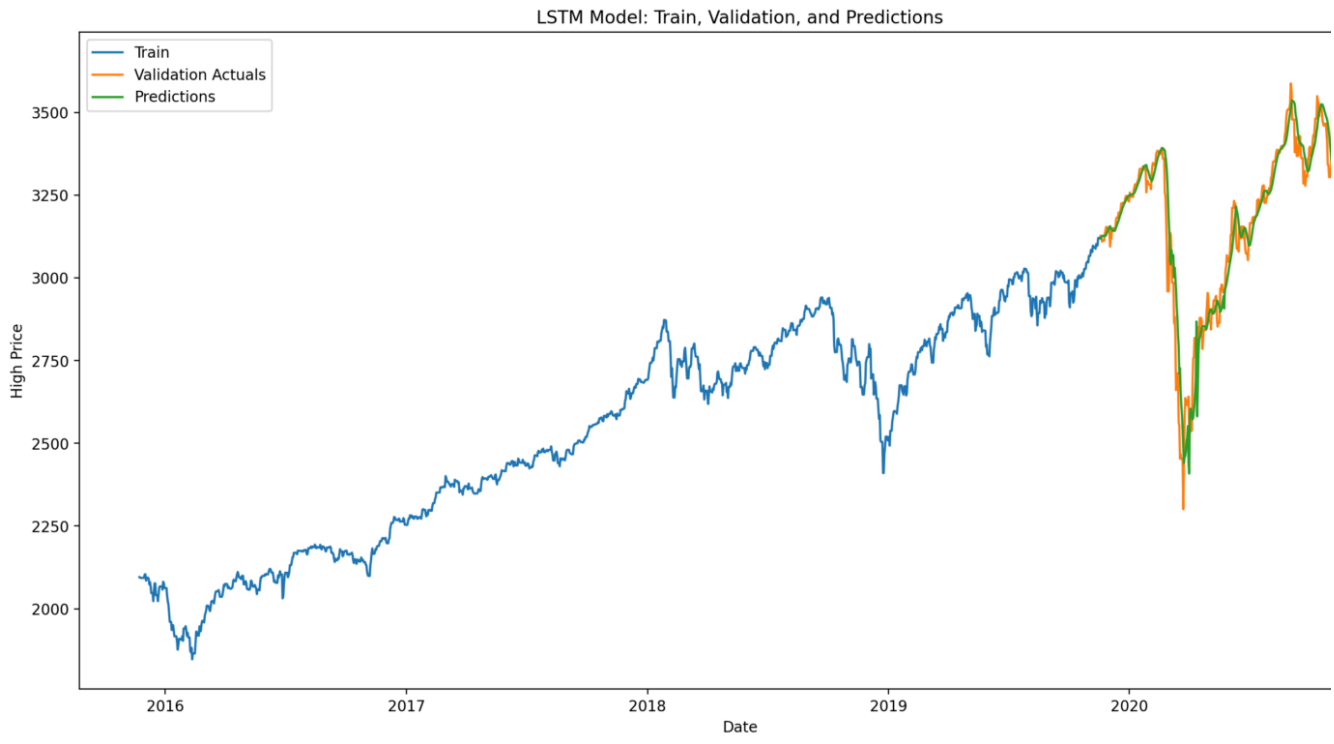
Train RMSE: 26.48

Validation RMSE: 74.01

Validation Predictions vs. Actuals



Train, Validation, and Predictions Plot



LSTM Model Performance Across Different Window Sizes

Processing window size: 40

Window size analysis complete!

window	train_rmse	val_rmse
5	25.4084	64.8065
8	27.1368	76.2343
10	32.2035	83.6342
15	24.4512	159.763
20	59.617	166.9709
30	54.0023	406.4411
40	63.5892	195.613

Complex LSTM Model Evaluation

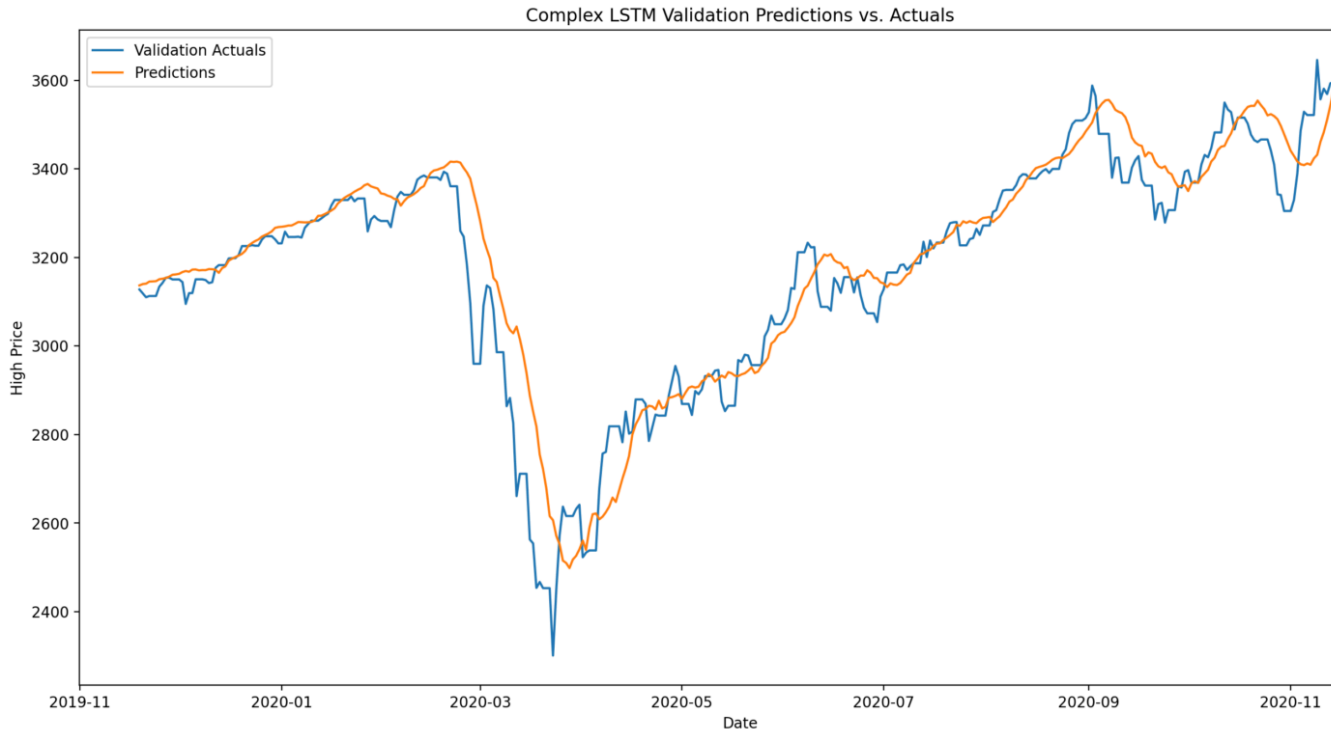
```
Training Complex LSTM Model...
```

Complex LSTM model trained!

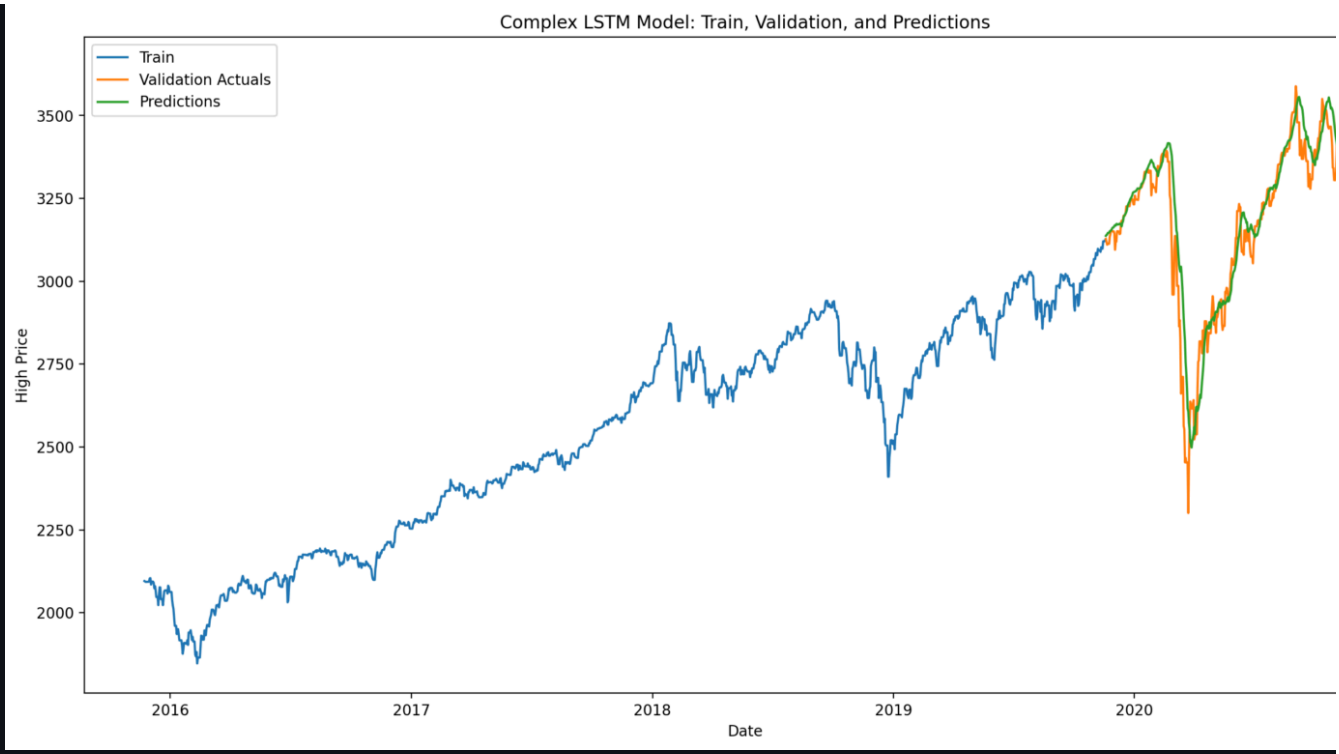
Train RMSE (Complex LSTM): 38.80

Validation RMSE (Complex LSTM): 90.92

Complex LSTM Validation Predictions vs. Actuals



Complex LSTM: Train, Validation, and Predictions Plot



Complex LSTM with Dropout Evaluation (10 runs)

Running dropout LSTM model 10/10

Dropout LSTM evaluation complete!

Train RMSEs with Dropout (10 runs):

```
[
  0 : 93.19
  1 : 37.71
  2 : 29.62
  3 : 102.69
  4 : 140.4
  5 : 76.16
  6 : 45.97
  7 : 63.08
  8 : 31.45
  9 : 40.8
]
```

Mean Train RMSE: 66.11

Standard Deviation Train RMSE: 36.65

Validation RMSEs with Dropout (10 runs):

```
[
  0 : 146.15
  1 : 101.98
  2 : 78.99
  3 : 153.2
  4 : 203.9
  5 : 139.09
  6 : 110.45
  7 : 111.24
```

```
  8 : 86.79
  9 : 98.35
]
```

Mean Validation RMSE: 123.01

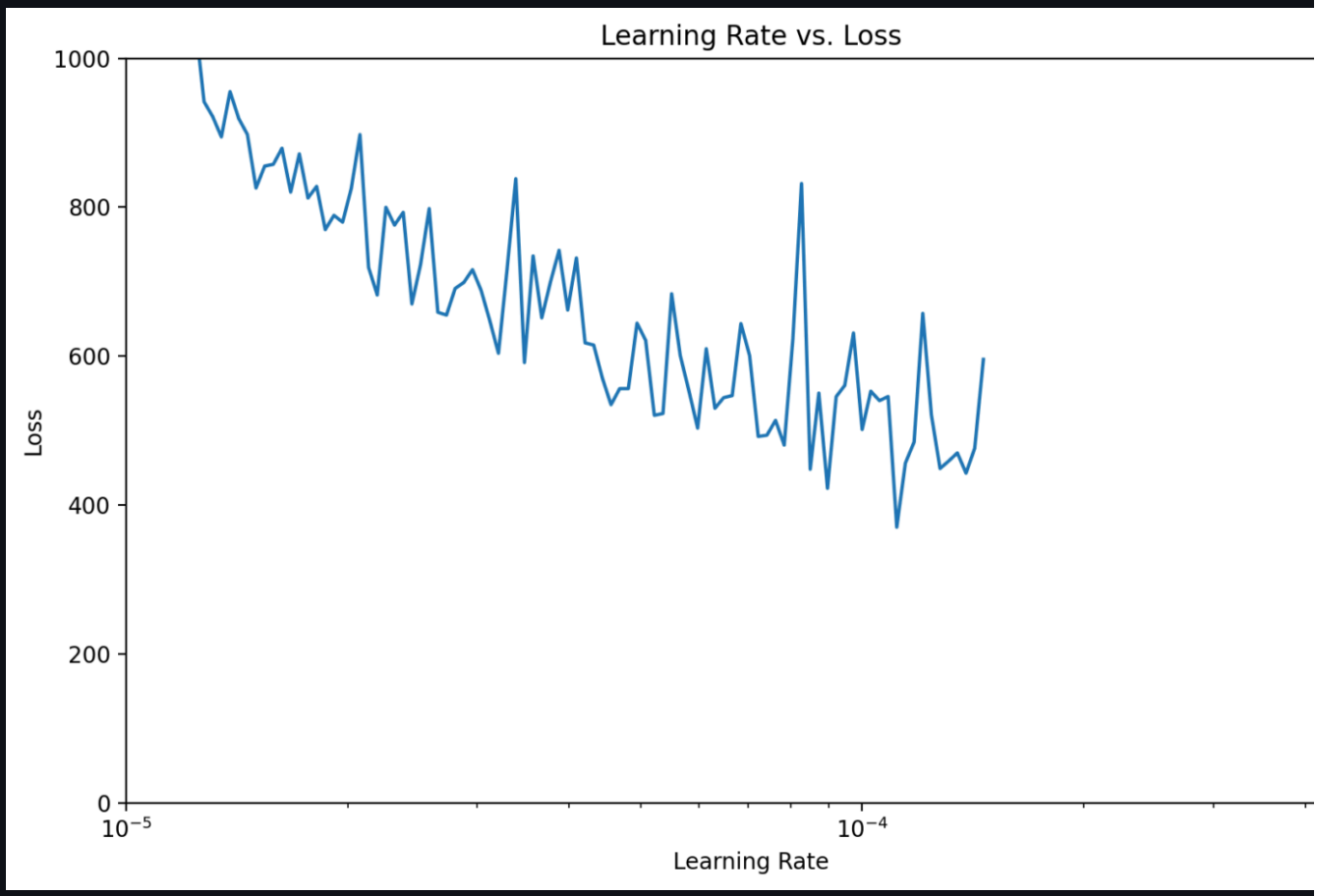
Standard Deviation Validation RMSE: 37.78

SimpleRNN Model with Learning Rate Finder

```
Running Learning Rate Finder for SimpleRNN...
```

Learning Rate Finder complete!

Learning Rate vs. Loss for SimpleRNN



Optimized SimpleRNN Model Evaluation

Running optimized SimpleRNN model 10/10

Optimized SimpleRNN evaluation complete!

Train RMSEs (Optimized SimpleRNN - 10 runs):

```
[
  0 : 17.62
  1 : 24.68
  2 : 17.34
  3 : 14.48
  4 : 18.49
  5 : 24.67
  6 : 19.67
```

```
  7 : 14.34
  8 : 17.58
  9 : 14.45
]
```

Mean Train RMSE: 18.33

Standard Deviation Train RMSE: 3.80

Validation RMSEs (Optimized SimpleRNN - 10 runs):

```
[
  0 : 41.68
  1 : 45.27
  2 : 39.68
  3 : 38.58
  4 : 41.15
  5 : 47.73
  6 : 46.05
  7 : 156.95
  8 : 40.18
  9 : 38.75
]
```

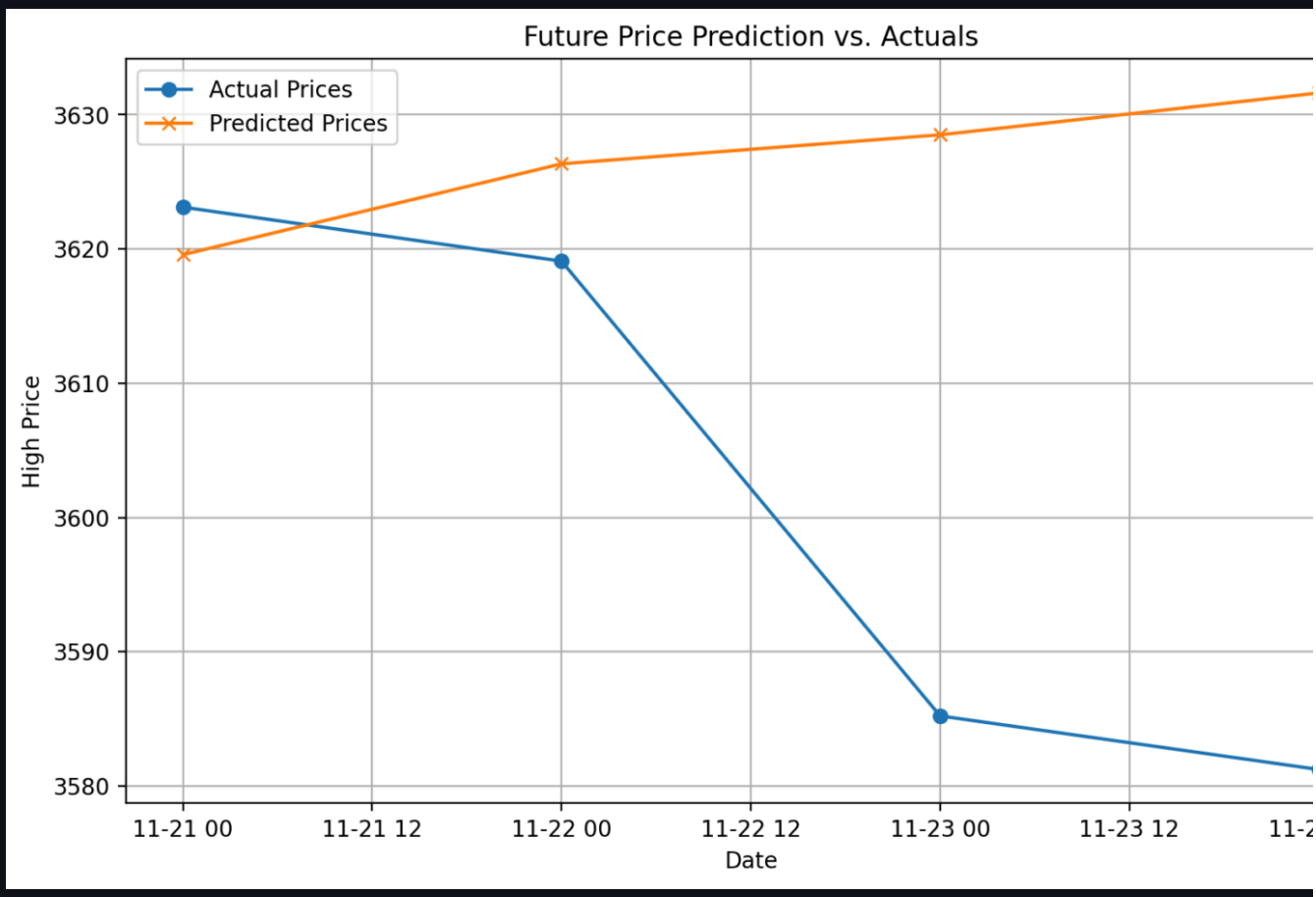
Mean Validation RMSE: 53.60

Standard Deviation Train RMSE: 36.45

Optimized SimpleRNN Validation Predictions vs. Actuals

Future Price Prediction

Predicted prices for the next 4 days: [3619.570068359375, 3626.335693359375, 3628.4970703125, 3631.614501953125]

Comparison of Actual and Predicted Future Prices

	Actual values	Predicted values
2020-11-21 00:00:00	3,623.1101	3,619.5701
2020-11-22 00:00:00	3,619.0901	3,626.3357
2020-11-23 00:00:00	3,585.22	3,628.4971
2020-11-24 00:00:00	3,581.23	3,631.6145